Monitoring Gas Ratio and Temperature Data for Structural Health Monitoring

Bilkent University

Electrical & Electronics Engineering Department

EEE 493 Industrial Design Project

Project Proposal

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Table of Contents

Table of Figures 3

1. Project Summary 4

2. Motivation 5

3. Novelty of the Project 6

4. Methodology 7

5. Expected Results 11

6. Equipment List 11

7. Work-Time Schedule 12

8. References 13

Table of Figures

Figure 1: WSN Model for Mines 7

Figure 2: DSR Algorithm 9

Figure 3: Jennic JN5168 10

Figure 4: Raspberry Pi 10

Figure 6: Equipment List Table 11

Figure 5: Gantt Chart of Work - Time Schedule 12

1. Project Summary

The project idea is based on designing and implementing a structural health monitoring system by creating a Wireless Sensor Network (WSN). The general idea for structural health monitoring in civil engineering is monitoring health status and behavior of a structure such as a bridge or building based on measured data by means of typical sensors (tilt, temperature etc.). As a different point of view, the main target of our project is to detect unexpected conditions that could be dangerous in mines instantaneously or even before they occur by designing appropriate system by using temperature and gas sensors. WSN can present a good functionality at a low cost for this type of application. Essentially, a well-established ad-hoc network is necessary to make connectivity and data collection reliable. To reach that goal the process is planned as: Firstly, a specific protocol will be written based on project`s requirements. Then, a software simulation program will be developed and the protocol will be implemented on this program. For several negative scenarios, unit tests will be performed in the simulation program. After that, the hardware of the system will be designed and the protocol will be embedded to the board. Later, the improvements for the quality will be analyzed and alarm system will be added. As a result, it is expected that if there is a serious temperature or gas change in the mine, then the system would alert the control unit immediately and enable workers to take precautions fast. In the future, this project can also be used as mobile by placing sensor nodes on workers so that the location of each worker could be seen easily for any accidents.

2. Motivation

As we all know, in recent years, there have been a lot of mine accidents in our country. A huge number of workers have passed away due to these accidents. These sad events have brought the safety issue in mines into question. In our opinion, one of the possible solutions is to monitor the structural health of mines continuously. For this purpose, we are aiming to develop a wireless sensor network (WSN) to monitor changes in characteristics of mines. By using our system in mines of our country, we will be able to detect possible dangers instantaneously, and this system will provide early warning system.

There are lots of publications and researches about wireless sensor networks, as this is an emerging technology. However, there are not any standards yet, and there is not a best application that provides a fine performance with a relatively low cost. Therefore, commercial wireless sensor network platforms are not very common. People mostly prefer to create their own WSN modules rather than purchasing it; because it is more convenient and less costly. Besides, there is not any commercial wireless sensor network product produced in Turkey. In the end of this project, we aim to meet this need and develop a ready to use product. With the use of our end product in mines of our country, we sincerely hope to assure our miners' well-being.

The only harmful side of this project is that when we set up the system in a mine, miners will be exposed to radio waves continuously. However, this negative effect would be negligible; as the system will be protecting the workers from more vital dangers.

The company that we worked with is KAREL Electronics Cooperation which is one of the largest enterprise telecommunication providers in Europe. In Bilkent Cyberpark, they have a R&D Center where product development activities are performed [1]. So, the possible use of the

project outcome for the company would be indoor places which are sensitive to temperature and/or gas changes. They have already implemented WSNs to patients in the hospitals and also developed vehicle tracking systems. So, they can also improve this project by adding different sensors.

3. Novelty of the Project

Wireless sensor network technology is such an area that there are still ongoing researches about it. However, there are just a few commercial products that are available in the worldwide market. One of those products which is called "Firefly Real-Time Sensor Networks-Coal Mine Monitoring" was developed to monitor the mine structure, like our project. It has light, motion, audio, temperature and acceleration sensors. Therefore, it provides utility monitoring, surveillance, location tracking and voice communication services for mines [2]. These features can be considered as factors that make the project superior. However, all these features in a single network will make it excessively expensive. Besides, a product that monitors the structural health of mine should include a gas detection sensor.

In our project, rather than using too many different kinds of sensors like motion, audio etc., we will have gas and temperature sensors which will help us monitor more crucial factors. Consequently, this will make our system cost efficient. In addition to this, in our country, there are no commercially available wireless sensor network systems for mine monitoring.

There are some existing WSN products which have obtained patents and one of them is the mentioned Firefly Real-Time Sensor Networks project. Since we have proposed a different solution to this problem, we think that we also have a chance to obtain a patent.

4. Methodology

As we mentioned earlier, the main problem is to monitor the structural health of mines in our country due to the fact that there has been many accidents in recent years causing many of our citizens' death. Our project's main idea is to propose a solution for this issue. We decided to construct a wireless sensor network (WSN) which will provide us a system to monitor the instantaneous changes in the environment. Wireless sensor networks consist of many individual nodes and a gateway which has the ability to connect to the Internet, or GSM network. Each node has its own sensors to read desired data from the environment. Gateway is actually the pool that information read from the nodes would be collected in it.

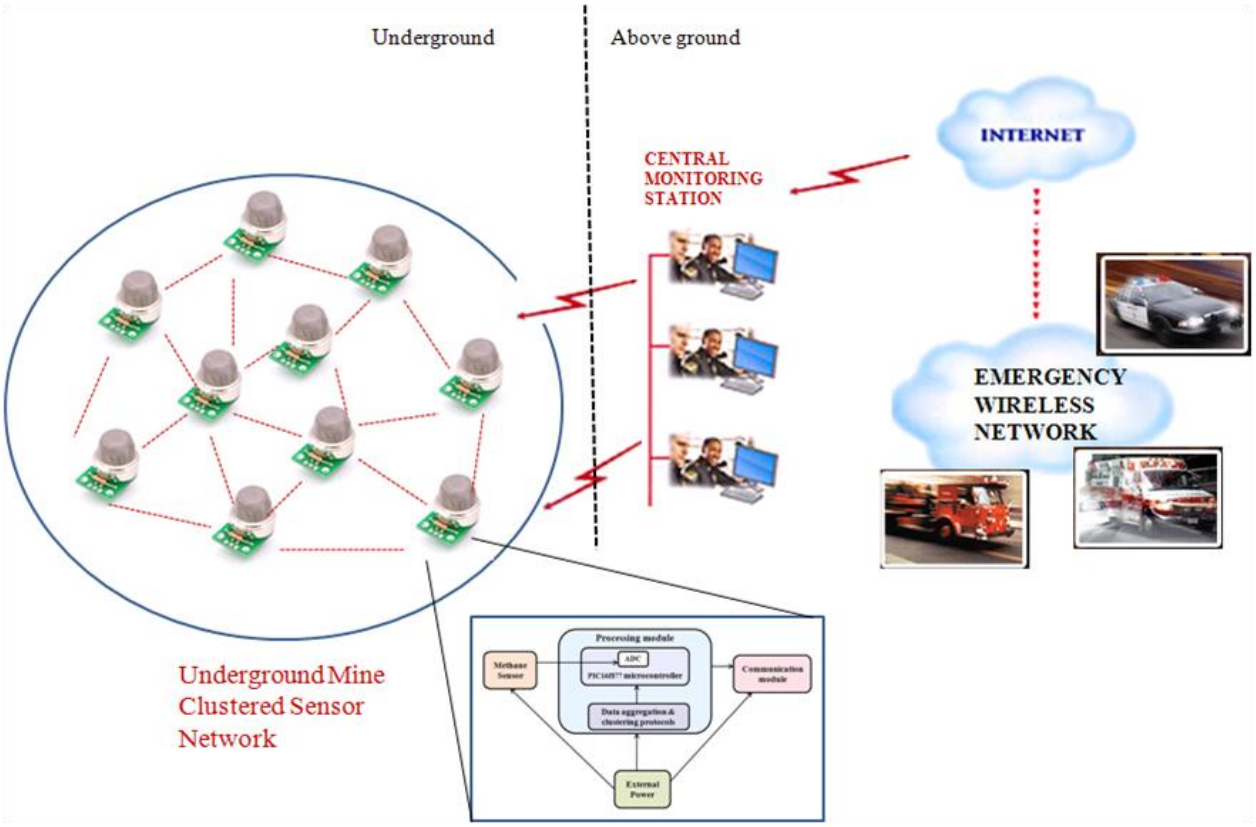


Figure 1: WSN Model for Mines

There exist mainly two parts for construction process of WSN. The first thing to do is completing the structure of network. After that, the system starts to carry information packages from each node to the gateway. Both of these operations need to be well-organized to avoid from disorders that might happen in the system. Therefore, all the nodes and data flow should be under the control of other nodes and mainly the gateway. For instance, in case of an inactivation of a node, the system should be able to detect the absence of this node and rearrange the related routes. All these process requires a well-defined protocol to realize and maintain the data flow.

We are planning to develop a Dynamic Source Routing (DSR) based protocol. DSR is kind of an on-demand algorithm and it has route discovery and route maintenance phases. The route discovery part contains two messages: *route request*, which is used for route determination, and *route reply*, which enables the source to learn and keep the complete route from source to destination. Route maintenance phase uses *route error* and *acknowledgements* packets. Acknowledgements are received from the neighboring nodes when data packets have been transmitted to the next hop and they are used as insurance for data transfer. In case of a problem that might occur in transmission process, which can be understood by not getting an acknowledgement packet, the route error packet is generated to report this problem [3]. This is, of course, the basic of the algorithm that we will use. We will improve this algorithm to get a full extent protocol.

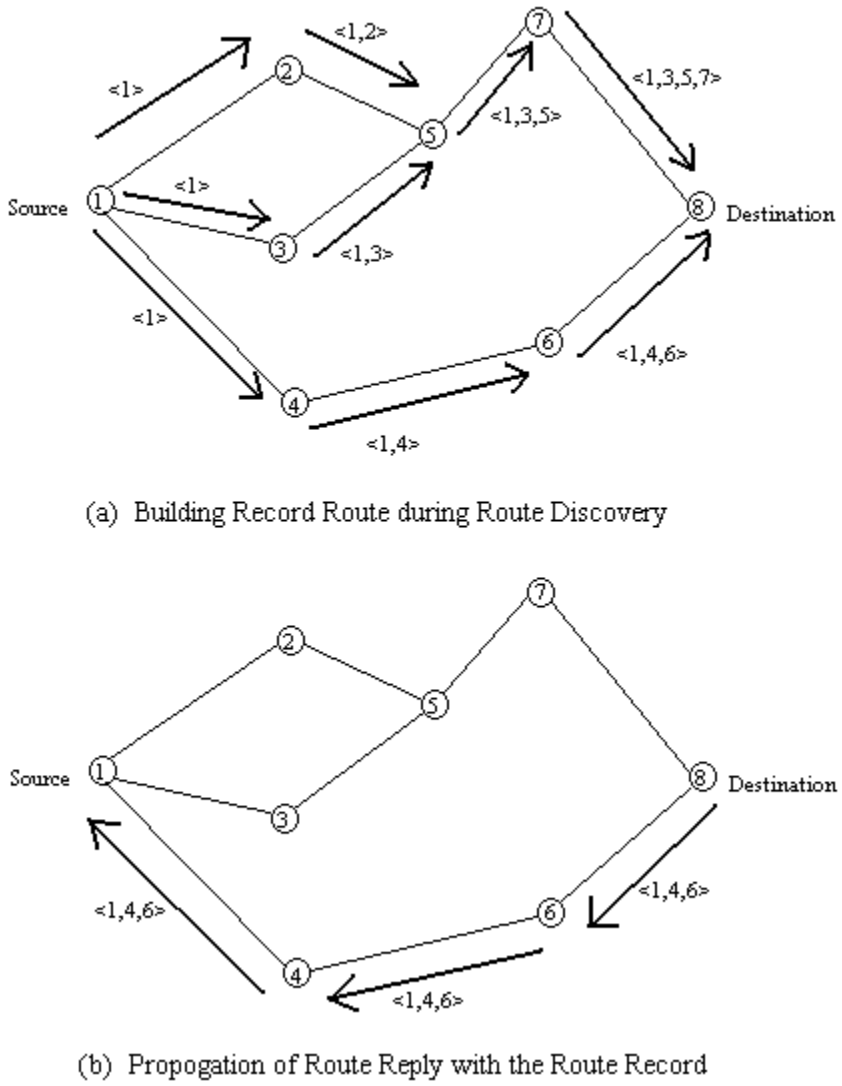


Figure 2: DSR Algorithm

The improvement process will be divided into two sections, which are construction of the complete protocol and its optimization period. The overall structure of protocol with necessary details will be decided on paper. Then the simulation part will take place and we will create a simulation environment for Wireless Sensor Network. Since there already exist some simulation software, like OMNeT++, we will be making use of them. If necessary, the simulation software

code will be composed on QT Creator. After getting the fully functional simulator program for WSN with our own protocol, we will constitute a network with at least 60 nodes and examine all possible routing cases. Then, we will make a comprehensive analysis of the results and try to optimize the algorithm. Also, overcome probable issues that we may encounter.

The developed protocol need to be introduced to each node in order to achieve a real life system. This protocol will be embedded into the hardware of nodes with C++ programming language. The hardware that we will use is Jennic Microcontroller as sensor nodes and Raspberry Pi as gateway.

Figure 3: Jennic JN5168

Figure 4: Raspberry Pi

Since these microcontrollers have their own RF transmission modules, they fit for our wireless system. Temperature and gas sensors will be used in each node, since our aim is to monitor the mine atmosphere. The design of the electronic board containing Jennic microcontroller and the sensors will also be our duty and we are planning to use Proteus ARES for this purpose. For the first phase, we are planning to have 6 nodes and 1 gateway. After programming all the nodes and gateway, they will start to communicate with each other and the

real life network will be established. At this point, we will need to test the network in an environment which can be considered as close as mines, like underground parking lots. The following step will be increasing the number of nodes, or maybe gateway, if the established network can communicate properly and provide the data flow successfully.

5. Expected Results

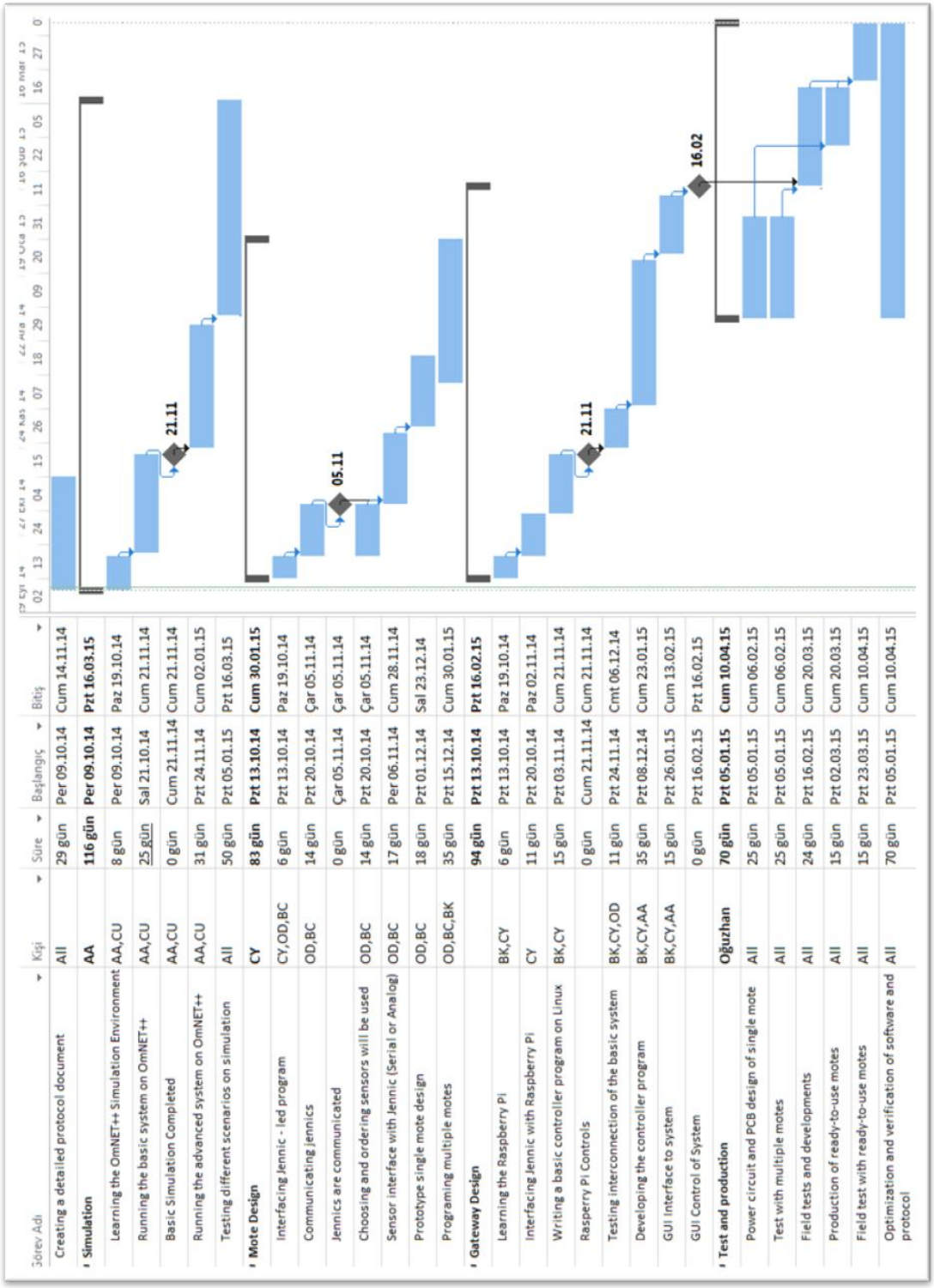
In the end of our project progress, we are expecting to have a real time working wireless sensor network. Sensors will be measuring data continuously and these data will be read at a rate of 1 KHz. The system will give information in case of a 1 degree (C) change in temperature. It is known that the carbon monoxide ratio in air starts getting dangerous for human health when it reaches above 100 ppm [4]. Therefore, a warning will be given when the carbon monoxide ratio is getting closer to dangerous levels. Rapid changes in gas and temperature level will be detected again by an alarm system.

6. Equipment List

Device	Part Number	Quantity (Planning)	How We Obtain	Unit Cost	Total Cost
Wireless Microcontroller Module	JN5148-001-M03	50	By purchase (also from company)	$16	$800
Raspberry Pi Model B+	RW-SF-12994	1	By purchase	$50	$50
Battery for motes		50	By purchase	$6	$300
Temperature sensor	LMT85	50	By purchase	$2	$100
CO and CH_4 gas sensor	MQ-9	50	By purchase	$2	$100
				Total Cost:	$1350

Figure 5: Equipment List Table

7. Work-Time Schedule



Görev Adı	Kişi	Süre	Başlangıç	Bitiş
Creating a detailed protocol document	All	29 gün	Per 09.10.14	Cum 14.11.14
Simulation	**AA**	**116 gün**	**Per 09.10.14**	**Pzt 16.03.15**
Learning the OmNET++ Simulation Environment	AA,CU	8 gün	Per 09.10.14	Paz 19.10.14
Running the basic system on OmNET++	AA,CU	25 gün	Sal 21.10.14	Cum 21.11.14
Basic Simulation Completed	AA,CU	0 gün	Cum 21.11.14	Cum 21.11.14
Running the advanced system on OmNET++	AA,CU	31 gün	Pzt 24.11.14	Cum 02.01.15
Testing different scenarios on simulation	All	50 gün	Pzt 05.01.15	Pzt 16.03.15
Mote Design	**CY**	**83 gün**	**Pzt 13.10.14**	**Cum 30.01.15**
Interfacing Jennic - led program	CY,OD,BC	6 gün	Pzt 13.10.14	Paz 19.10.14
Communicating jennics	OD,BC	14 gün	Pzt 20.10.14	Çar 05.11.14
Jennics are communicated		0 gün	Çar 05.11.14	Çar 05.11.14
Choosing and ordering sensors will be used	OD,BC	14 gün	Pzt 20.10.14	Çar 05.11.14
Sensor interface with Jennic (Serial or Analog)	OD,BC	17 gün	Per 06.11.14	Cum 28.11.14
Prototype single mote design	OD,BC	18 gün	Pzt 01.12.14	Sal 23.12.14
Programing multiple motes	OD,BC,BK	35 gün	Pzt 15.12.14	Cum 30.01.15
Gateway Design		**94 gün**	**Pzt 13.10.14**	**Pzt 16.02.15**
Learning the Raspberry Pi	BK,CY	6 gün	Pzt 13.10.14	Paz 19.10.14
Interfacing Jennic with Raspberry Pi	CY	11 gün	Pzt 20.10.14	Paz 02.11.14
Writing a basic controller program on Linux	BK,CY	15 gün	Pzt 03.11.14	Cum 21.11.14
Rasperry Pi Controls		0 gün	Cum 21.11.14	Cum 21.11.14
Testing interconnection of the basic system	BK,CY,OD	11 gün	Pzt 24.11.14	Cmt 06.12.14
Developing the controller program	BK,CY,AA	35 gün	Pzt 08.12.14	Cum 23.01.15
GUI Interface to system	BK,CY,AA	15 gün	Pzt 26.01.15	Cum 13.02.15
GUI Control of System		0 gün	Pzt 16.02.15	Pzt 16.02.15
Test and production	**Oğuzhan**	**70 gün**	**Pzt 05.01.15**	**Cum 10.04.15**
Power circuit and PCB design of single mote	All	25 gün	Pzt 05.01.15	Cum 06.02.15
Test with multiple motes	All	25 gün	Pzt 05.01.15	Cum 06.02.15
Field tests and developments	All	24 gün	Pzt 16.02.15	Cum 20.03.15
Production of ready-to-use motes	All	15 gün	Pzt 02.03.15	Cum 20.03.15
Field test with ready-to-use motes	All	15 gün	Pzt 23.03.15	Cum 10.04.15
Optimization and verification of software and protocol	All	70 gün	Pzt 05.01.15	Cum 10.04.15

Figure 6: Gantt Chart of Work - Time Schedule

8. References

[1]. Karel Electronics, “R&D / Technology,” http://www.karel-electronics.com/rd-technology, Accessed October 8, 2014.

[2]. Firefly Sensor Networks, “Real-Time Wireless Sensor Network Platform,” http://www.ece.cmu.edu/firefly/index.html, Accessed October 10, 2014.

[3]. A. Boukerche, “Routing protocols in ad hoc networks,” *Computer Networks*, vol. 55, issue 11, pp. 3032-3080, 2011.

[4]. Prockop LD, Chichkova RI, "Carbon Monoxide Intoxication: An Updated Review," *Journal of the Neurological Sciences*, vol. 262, pp. 122–130, 2007.